

# Infosys SP/DSE Must-Solve Problems

 **Based on 2025-2026 Test Patterns & Frequently Asked Questions**

 **Exam Structure (Confirmed)**

- **Duration:** 3 hours
  - **Questions:** 3 problems
  - **Difficulty Split:**
    - 1 Easy (Arrays/Strings/Hash Maps/Basic DS)
    - 1-2 Medium (Recursion/Backtracking/Logic)
    - ~~1 Hard (Skip DP/Graphs)~~
  - **Languages:** Java, Python, C++, C
  - **Target:** Solve 2/3 correctly to pass
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## HIGHEST PRIORITY - Arrays & Strings (70% Chance!)

### Arrays - Easy (MUST DO - Practice 2 times each)

| #  | Problem                             | Platform | Difficulty | Pattern       | Priority |
|----|-------------------------------------|----------|------------|---------------|----------|
| 1  | Two Sum                             | LC 1     | Easy       | Hash Map      | ★★★★★    |
| 2  | Second Largest Element              | GFG      | Easy       | Single Pass   | ★★★★★    |
| 3  | Remove Duplicates from Sorted Array | LC 26    | Easy       | Two Pointers  | ★★★★★    |
| 4  | Best Time to Buy and Sell Stock     | LC 121   | Easy       | Kadane's      | ★★★★     |
| 5  | Move Zeros                          | LC 283   | Easy       | Two Pointers  | ★★★★     |
| 6  | Contains Duplicate                  | LC 217   | Easy       | Hash Set      | ★★★★     |
| 7  | Rotate Array                        | LC 189   | Easy       | Reverse Trick | ★★★★     |
| 8  | Find Missing Number                 | LC 268   | Easy       | Math/XOR      | ★★★      |
| 9  | Merge Sorted Array                  | LC 88    | Easy       | Two Pointers  | ★★★      |
| 10 | Leaders in Array                    | GFG      | Easy       | Right to Left | ★★★★     |

**Direct Link:** [LeetCode Easy Arrays](#)

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### Arrays - Medium (High Frequency in Infosys)

| #  | Problem                      | Platform | Difficulty | Pattern           | Priority |
|----|------------------------------|----------|------------|-------------------|----------|
| 11 | Product of Array Except Self | LC 238   | Medium     | Prefix/Suffix     | ★★★★★    |
| 12 | Maximum Subarray (Kadane's)  | LC 53    | Medium     | Kadane's          | ★★★★★    |
| 13 | Subarray Sum Equals K        | LC 560   | Medium     | Hash Map + Prefix | ★★★★★    |
| 14 | 3Sum                         | LC 15    | Medium     | Two Pointers      | ★★★★     |
| 15 | Sort Colors (Dutch Flag)     | LC 75    | Medium     | Three Pointers    | ★★★★     |
| 16 | Find All Duplicates          | LC 442   | Medium     | Index Marking     | ★★★      |
| 17 | Next Permutation             | LC 31    | Medium     | Logic             | ★★★      |

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## Strings - Easy (MUST DO)

| #  | Problem                       | Platform | Difficulty | Pattern           | Priority |
|----|-------------------------------|----------|------------|-------------------|----------|
| 18 | Valid Anagram                 | LC 242   | Easy       | Frequency Count   | ★★★★★    |
| 19 | Valid Palindrome              | LC 125   | Easy       | Two Pointers      | ★★★★★    |
| 20 | First Non-Repeating Character | GFG      | Easy       | Hash Map          | ★★★★★    |
| 21 | Reverse String                | LC 344   | Easy       | Two Pointers      | ★★★★     |
| 22 | Reverse Words in a String     | LC 151   | Medium     | Split/Join        | ★★★★     |
| 23 | Longest Common Prefix         | LC 14    | Easy       | Character by char | ★★★      |
| 24 | Valid Parentheses (Stack)     | LC 20    | Easy       | Stack             | ★★★★★    |

## Strings - Medium (Frequently Asked)

| #  | Problem                             | Platform | Difficulty | Pattern        | Priority |
|----|-------------------------------------|----------|------------|----------------|----------|
| 25 | Longest Substring Without Repeating | LC 3     | Medium     | Sliding Window | ★★★★★    |

| #  | Problem                       | Platform | Difficulty | Pattern       | Priority |
|----|-------------------------------|----------|------------|---------------|----------|
| 26 | Group Anagrams                | LC 49    | Medium     | Hash Map      | ★★★★★    |
| 27 | Longest Palindromic Substring | LC 5     | Medium     | Expand Center | ★★★★     |
| 28 | String to Integer (atoi)      | LC 8     | Medium     | Edge Cases    | ★★★      |

### HIGH PRIORITY - Hash Maps (60% Chance!)

| #  | Problem                       | Platform | Difficulty | Pattern       | Priority |
|----|-------------------------------|----------|------------|---------------|----------|
| 29 | Count Repeated & Non-Repeated | GFG      | Easy       | Frequency Map | ★★★★★    |
| 30 | Top K Frequent Elements       | LC 347   | Medium     | Hash + Heap   | ★★★★     |
| 31 | Longest Consecutive Sequence  | LC 128   | Medium     | Hash Set      | ★★★★     |
| 32 | Four Sum                      | LC 18    | Medium     | Hash Map      | ★★★      |

## MEDIUM PRIORITY - Linked Lists (40% Chance)

### Easy

| #  | Problem                | Platform | Difficulty | Pattern             | Priority  |
|----|------------------------|----------|------------|---------------------|-----------|
| 33 | Reverse Linked List    | LC 206   | Easy       | Iterative/Recursive | ★ ★ ★ ★ ★ |
| 34 | Merge Two Sorted Lists | LC 21    | Easy       | Two Pointers        | ★ ★ ★ ★ ★ |
| 35 | Linked List Cycle      | LC 141   | Easy       | Floyd's             | ★ ★ ★ ★ ★ |
| 36 | Middle of Linked List  | LC 876   | Easy       | Fast-Slow           | ★ ★ ★ ★   |
| 37 | Palindrome Linked List | LC 234   | Easy       | Reverse Half        | ★ ★ ★     |

### Medium

| #  | Problem                  | Platform | Difficulty | Pattern      | Priority  |
|----|--------------------------|----------|------------|--------------|-----------|
| 38 | Remove Nth Node from End | LC 19    | Medium     | Two Pointers | ★ ★ ★ ★ ★ |
| 39 | Linked List Cycle II     | LC 142   | Medium     | Floyd's      | ★ ★ ★ ★   |
| 40 | Add Two Numbers          | LC 2     | Medium     | Carry Logic  | ★ ★ ★ ★   |

| #  | Problem      | Platform | Difficulty | Pattern         | Priority |
|----|--------------|----------|------------|-----------------|----------|
| 41 | Reorder List | LC 143   | Medium     | Reverse + Merge | ★ ★ ★    |

## MEDIUM PRIORITY - Stacks & Queues (35% Chance)

| #  | Problem                          | Platform | Difficulty | Pattern         | Priority  |
|----|----------------------------------|----------|------------|-----------------|-----------|
| 42 | Valid Parentheses                | LC 20    | Easy       | Stack           | ★ ★ ★ ★ ★ |
| 43 | Implement Queue using Stacks     | LC 232   | Easy       | Two Stacks      | ★ ★ ★ ★ ★ |
| 44 | Implement Stack using Queues     | LC 225   | Easy       | Two Queues      | ★ ★ ★ ★   |
| 45 | Next Greater Element             | LC 496   | Easy       | Monotonic Stack | ★ ★ ★ ★ ★ |
| 46 | Min Stack                        | LC 155   | Medium     | Stack Design    | ★ ★ ★ ★   |
| 47 | Evaluate Reverse Polish Notation | LC 150   | Medium     | Stack           | ★ ★ ★ ★ ★ |
| 48 | Daily Temperatures               | LC 739   | Medium     | Monotonic Stack | ★ ★ ★     |

## **IMPORTANT - Recursion (50% Chance!)**

### **Basic Recursion (Must Know)**

| #  | Problem                    | Platform | Difficulty | Pattern           | Priority |
|----|----------------------------|----------|------------|-------------------|----------|
| 49 | Fibonacci Number           | LC 509   | Easy       | Basic Recursion   | ★★★★★    |
| 50 | Power(x, n)                | LC 50    | Medium     | Divide & Conquer  | ★★★★★    |
| 51 | Reverse String (Recursive) | LC 344   | Easy       | Basic Recursion   | ★★★★     |
| 52 | Sum of Digits              | GFG      | Easy       | Basic Recursion   | ★★★★     |
| 53 | Tower of Hanoi             | GFG      | Medium     | Classic Recursion | ★★★      |

## **BACKTRACKING - Medium Priority (30% Chance)**

| #  | Problem      | Platform | Difficulty | Pattern      | Priority |
|----|--------------|----------|------------|--------------|----------|
| 54 | Subsets      | LC 78    | Medium     | Backtracking | ★★★★★    |
| 55 | Permutations | LC 46    | Medium     | Backtracking | ★★★★★    |



| #  | Problem                   | Platform | Difficulty | Pattern             | Priority |
|----|---------------------------|----------|------------|---------------------|----------|
| 56 | Combination Sum           | LC 39    | Medium     | Backtracking        | ★★★★★    |
| 57 | Generate Parentheses      | LC 22    | Medium     | Backtracking        | ★★★★     |
| 58 | Letter Combinations Phone | LC 17    | Medium     | Backtracking        | ★★★★     |
| 59 | Word Search               | LC 79    | Medium     | Backtracking + Grid | ★★★★     |
| 60 | N-Queens                  | LC 51    | Hard       | Backtracking        | ★★★      |

### **SPECIAL - Pattern & Logic Problems (Infosys Favorites!)**

These appear frequently in Infosys tests but not on LeetCode:

| #  | Problem                                        | Description                                   | Priority  |
|----|------------------------------------------------|-----------------------------------------------|-----------|
| 61 | <b>Print Diamond Pattern</b>                   | Print diamond using stars                     | ★ ★ ★ ★ ★ |
| 62 | <b>Count Repeated vs Non-Repeated</b>          | Array frequency analysis                      | ★ ★ ★ ★ ★ |
| 63 | <b>Smallest Power of 2 <math>\geq</math> N</b> | For each array element                        | ★ ★ ★ ★   |
| 64 | <b>Leaders in Array</b>                        | Elements greater than all to right            | ★ ★ ★ ★ ★ |
| 65 | <b>Max Difference (larger after smaller)</b>   | $\text{arr}[j] - \text{arr}[i]$ where $j > i$ | ★ ★ ★ ★   |

## 🎓 7-DAY PRACTICE SCHEDULE (Jan 12-18)

### Day 1 (Jan 12) - Arrays Foundation

**Morning:** Review array cheatsheet

**Practice:**

- Two Sum (LC 1) ★ ★ ★ ★ ★
- Remove Duplicates (LC 26) ★ ★ ★ ★ ★
- Move Zeros (LC 283) ★ ★ ★ ★

- Leaders in Array (GFG) ★ ★ ★ ★
- Second Largest Element (GFG) ★ ★ ★ ★ ★
- Contains Duplicate (LC 217) ★ ★ ★ ★
- Product Except Self (LC 238) ★ ★ ★ ★ ★

**Goal:** Master two-pointer and single-pass techniques

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## Day 2 (Jan 13) - Strings + Hash Maps

**Morning:** Review string & hash map syntax

**Practice:**

- Valid Anagram (LC 242) ★ ★ ★ ★ ★
- Valid Palindrome (LC 125) ★ ★ ★ ★ ★
- First Non-Repeating Char (GFG) ★ ★ ★ ★ ★
- Group Anagrams (LC 49) ★ ★ ★ ★ ★
- Longest Substring Without Repeating (LC 3) ★ ★ ★ ★ ★
- Count Repeated/Non-Repeated (GFG) ★ ★ ★ ★ ★
- Subarray Sum Equals K (LC 560) ★ ★ ★ ★ ★

**Goal:** Master frequency counting and sliding window

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### Day 3 (Jan 14) - Linked Lists

**Morning:** Review linked list patterns

**Practice:**

- Reverse Linked List (LC 206) ★ ★ ★ ★ ★
- Merge Two Sorted Lists (LC 21) ★ ★ ★ ★ ★
- Linked List Cycle (LC 141) ★ ★ ★ ★ ★
- Middle of Linked List (LC 876) ★ ★ ★ ★ ★
- Remove Nth from End (LC 19) ★ ★ ★ ★ ★
- Add Two Numbers (LC 2) ★ ★ ★ ★

**Goal:** Master Floyd's algorithm and reversal

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### Day 4 (Jan 15) - Stacks & Queues

**Morning:** Review stack/queue implementations

**Practice:**

- Valid Parentheses (LC 20) ★ ★ ★ ★ ★
- Next Greater Element (LC 496) ★ ★ ★ ★ ★
- Implement Queue using Stacks (LC 232) ★ ★ ★ ★ ★
- Implement Stack using Queues (LC 225) ★ ★ ★ ★
- Evaluate RPN (LC 150) ★ ★ ★ ★ ★
- Min Stack (LC 155) ★ ★ ★ ★

**Goal:** Master stack-based patterns

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## Day 5 (Jan 16) - Recursion & Backtracking

**Morning:** Review recursion templates

**Practice:**

- Fibonacci (LC 509) ★ ★ ★ ★ ★
- Power(x, n) (LC 50) ★ ★ ★ ★ ★
- Subsets (LC 78) ★ ★ ★ ★ ★
- Permutations (LC 46) ★ ★ ★ ★ ★
- Combination Sum (LC 39) ★ ★ ★ ★ ★

- Generate Parentheses (LC 22) ★ ★ ★ ★

**Goal:** Understand backtracking template

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## **Day 6 (Jan 17) - Mixed Medium Problems**

**Practice:**

- Maximum Subarray (LC 53) ★ ★ ★ ★ ★
- Top K Frequent Elements (LC 347) ★ ★ ★ ★
- 3Sum (LC 15) ★ ★ ★ ★
- Longest Consecutive Sequence (LC 128) ★ ★ ★ ★
- Word Search (LC 79) ★ ★ ★ ★
- Daily Temperatures (LC 739) ★ ★ ★

**Goal:** Practice problem identification

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## **Day 7 (Jan 18 Morning) - MOCK TEST + REVISION**

**Morning (8-11 AM):**

- **Mock Test 1:** Pick any 3 random problems (1 easy, 2 medium) - 90 minutes
  - Example: Two Sum + Group Anagrams + Reverse Linked List
- **Mock Test 2:** Different 3 problems - 90 minutes
  - Example: Valid Parentheses + Subarray Sum K + Permutations

### Afternoon (12-3 PM):

- Review all ★ ★ ★ ★ ★ problems
- Practice pattern problems (Diamond, Leaders)
- Relax, stay calm

**Evening:** Light review, sleep early

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### **INFOSYS-SPECIFIC TIPS**

#### **What Infosys LOVES to Ask:**

1. **Frequency counting problems** (Hash Maps)
2. **Anagram-related questions** (Strings)
3. **Array manipulation** (Two pointers, Kadane's)

4. **Pattern printing** (Loops)
5. **Basic recursion** (Fibonacci, factorial variants)
6. **Stack-based validation** (Parentheses, expressions)

### **Common Mistakes to Avoid:**

- ❌ Not handling edge cases (empty arrays, single element)
- ❌ Using `==` for string comparison (use `.equals()`)
- ❌ Forgetting to check `null` inputs
- ❌ Not testing with negative numbers
- ❌ Off-by-one errors in loops
- ❌ Not optimizing after brute force

### **During the Exam:**

1. **Read ALL 3 questions first** (2-3 minutes)
2. **Identify the easiest one** - solve it first!
3. **For each problem:**
  - Understand input/output format carefully
  - Think of edge cases



- Write approach in comments first
  - Code step by step
  - Test with examples before submitting
4. **Time split:** 45-50 min per problem max
  5. **If stuck > 40 min:** Move to next problem
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## RECOMMENDED PRACTICE PLATFORMS

1. **LeetCode** - For standard problems (Easy | Medium)
  2. **GeeksforGeeks** - For Infosys-specific patterns (GFG Practice)
  3. **HackerRank** - Similar interface to Infosys test (Arrays)
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## FINAL CHECKLIST (Jan 17 Night)

- ☐ Solved at least 40+ problems from above list
- ☐ Can implement all data structures from scratch
- ☐ Know time/space complexity of all patterns
- ☐ Completed 2 full mock tests (3 problems in 3 hours)

- ☐ Comfortable with Java Collections (HashMap, ArrayList, etc.)
  - ☐ Tested edge cases in all practice problems
  - ☐ Reviewed all ★ ★ ★ ★ ★ problems one last time
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 **YOU'VE GOT THIS!**

**Remember:**

- Infosys wants to see **clean, working code**
- **2/3 correct = You Pass!**
- Focus on arrays, strings, and hash maps first
- Stay calm, read carefully, test your code
- You've prepared well - trust yourself!

**Good luck on Jan 18!** 💪 🔥